CHANDU PRASAD BHAIRAPU



AI Evaluation Engineer • LLM Evaluation Pipelines, AI Agents, Python

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ABOUT ME

AI engineer with an MSc in Artificial Intelligence and 3 years of professional AI/ML experience, most recently building LLM evaluation pipelines and the multi-agent systems they measure. At an early-stage startup I own a multi-agent extraction system end-to-end and measure every iteration against an SAP-derived ground truth (precision/recall from 33%/45% to 93%/90%); for a high-volume source I built deterministic code-based extraction instead of LLM calls, cutting a $200,000 projected LLM budget to near zero. My Master's thesis automated RAG retrieval and generation evaluation and benchmarked embedding models on 1M+ chunks. I work in Python with LangGraph, LangChain, Docker, CI/CD, AWS and Azure.

EXPERIENCE

Associate Generative AI Developer - Agentic Indexing Pipeline Engineer for German Energy Sector

Kniit GmbH

November 2025 – present Muldestausee, Germany

- Designed the multi-agent LangGraph extraction workflow (entity-extraction, relationship-analysis, consolidation, and reflection agents running in parallel) for complex regulatory and technical documents, and its evaluation: scored outputs against a ground-truth dataset derived from SAP systems and measured precision/recall on each prompt and agent iteration, driving it from 33%/45% to 93%/90%.
- Sole developer owning the knowledge-extraction pipeline end-to-end (data ingestion, chunking, prompt engineering, and indexing with agentic, deterministic, and monolithic strategies), reporting directly to the CTO and working with SAP domain experts across the German energy sector.
- Engineered a deterministic, code-based extraction pipeline instead of LLM calls for a second, high-volume source stream, cutting a $200,000 projected LLM budget to near-zero recurring cost and pushing accuracy toward 100%.
- Designed and implemented a Model Context Protocol (MCP) query service with millisecond-latency retrieval from the knowledge graph, integrated into Claude Code for agentic development workflows.
- Developed Expert Interface, a graph curation web app (Python microservices backend, React frontend) on Azure, where domain experts validate and approve AI-extracted knowledge-graph changes.
- **Tech stack: Python, LangGraph, LangChain, LiteLLM, MCP, RAG, GraphRAG, Vector Search, FastAPI, Neo4j, MongoDB, Postgres, Docker, Kubernetes, Azure, React**

GenAI Developer (AI Consulting Intern) - Development and Deployment of an Agentic AI Workflow

Covestro Deutschland AG

December 2024 – April 2025 Leverkusen, Germany

- Designed and deployed an agentic AI application with LangGraph that routes each request to vector database retrieval, general inference, or summarization.
- Built the Python backend and REST APIs with FastAPI, with API testing in Postman, plus a React frontend; containerized the stack with Docker and deployed it to AWS using CDK, with automated GitLab CI/CD pipelines.
- **Tech stack: Python, LangGraph, LangChain, FastAPI, Postman, GitLab CI/CD, Docker, AWS (CDK, EC2, Lambda), React**

GenAI Developer (Master's Thesis) - Evaluating the Performance of RAG Systems

Covestro Deutschland AG

May 2024 – November 2024 Leverkusen, Germany

- Automated evaluation of RAG generation and retrieval with RAGAS and PytrecEval, scoring model answers against chemist-provided ground truth.
- Benchmarked multilingual embedding models on over 1M data chunks of Covestro's internal chemical data and used the results to set chunking, indexing, and retrieval for the RAG pipeline.
- Built the evaluation infrastructure for PrismaChat on AWS (Lambda, SQS, DynamoDB) with CDK.
- **Tech stack: Python, RAGAS, PytrecEval, LangChain, Hugging Face Transformers, Vector Databases, Azure AI Search, AWS (Lambda, SQS, DynamoDB, S3, CDK), Docker, PyTorch**

Data Scientist (Intern) – ML-Based Prediction of Prepolymer Synthesis Outcomes

Covestro Deutschland AG

November 2023 – April 2024 | Leverkusen, Germany

- Built and compared Random Forest, XGBoost, and LSTM models to predict prepolymer synthesis outcomes (e.g. viscosity) from time-series process data, served to chemists through a Streamlit dashboard; improved early identification of batches likely to deviate from target specification.
- **Tech stack: Python, Scikit-Learn, XGBoost, TensorFlow/Keras, Pandas, Streamlit**

Machine Learning Engineer - Electrical Vehicle Load Forecasting using Time Series Analysis

Cognizant Technology Solutions

September 2021 – March 2022 | Hyderabad, India

- Developed ARIMA, SARIMA, XGBoost, FB Prophet, and LSTM models for EV load forecasting; the LSTM reached 96% test accuracy.
- **Tech stack: Python, Scikit-Learn, TensorFlow**

PROJECTS

Project: Agent Eval Harness | Python | pytest | GitHub Actions | OpenTelemetry | Arize Phoenix

- Evaluation pipeline for a tool-calling LangGraph agent: golden datasets for regression testing (edge cases, synthetic multi-turn scenarios); deterministic checks run first (JSON schema validation of structured outputs and function calling payloads, regular expressions, API status checks, embedding similarity), and LLM-as-a-judge scores only what they cannot decide.
- Statistical evaluation runs each case several times and reports pass rates with confidence intervals; adversarial testing covers prompt injection, malformed inputs, unexpected tool responses, and tool-calling loop limits.
- Runs in GitHub Actions with caching of LLM responses and mocking of tools to cut API calls; observability and tracing via OpenTelemetry and Arize Phoenix; asynchronous load testing of the non-deterministic endpoint.

TECHNICAL SKILLSET

- **AI Evaluation:** LLM evaluation pipelines, ground-truth datasets, precision/recall, RAG evaluation (RAGAS, PytrecEval), embedding model benchmarking, deterministic (code-based) pipelines in place of LLM calls
- **Python & APIs:** Python, FastAPI, REST APIs, API testing (Postman), Python microservices, SQL
- **AI Agents & LLMs:** LangGraph (multi-agent workflows), LangChain, LiteLLM, MCP, prompt engineering, Hugging Face Transformers, RAG, GraphRAG
- **Embeddings & Databases:** embedding models, vector databases (Qdrant, Pinecone, Azure AI Search), vector search, Neo4j, MongoDB, Postgres, DynamoDB
- **CI/CD & Cloud:** GitLab CI/CD, Git, Docker, Kubernetes, AWS (CDK, Lambda, SQS, EC2, S3, VPC), Azure
- **AI Development Tools:** Claude Code (custom skills, agentic coding workflows, MCP integration)
- **Machine Learning:** Scikit-Learn, PyTorch, TensorFlow, XGBoost, time-series analysis

EDUCATION

Master of Science in Artificial Intelligence — March 2022 – April 2025
Technische Hochschule Würzburg-Schweinfurt, Würzburg, Germany

Bachelor of Technology in Mechanical Engineering, Minor in Computer Science & Engineering — August 2017 – July 2021
G Pulla Reddy Engineering College, Kurnool, India

PUBLICATIONS & CERTIFICATIONS

- **[1]** Navigating the Future: An approach of Autonomous Indoor Vehicles (conference publication)
- Udemy - LLMs Mastery : Transformers & Generative AI

LANGUAGES

English (Business Fluent) | German (B1) | Hindi (Fluent) | Telugu (Native)